

Long Quan Ton

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EDUCATION

University of Technology Sydney & HCMUT

Bachelor of Artificial Intelligence; GPA: 6.75/7.0

Ho Chi Minh City, Vietnam

Expected 2028

EXPERIENCE

Data Annotator (Part-time, Remote)

Data Quality Control & Labeling

GlobalTech VietNam

May 2026 – Present

- **Data Processing & Annotation:** Accurately processed and annotated complex datasets across multiple structured and unstructured formats, primarily working with HTML, JSON, and Excel files.
- **Quality Assurance:** Conducted rigorous quality assurance checks and precise data labeling, consistently meeting strict weekly page-volume targets while maintaining high data integrity.

LEADERSHIP & ACTIVITIES

Weatherise – Top 10 Finalists Vietnam AI Open Hackathon

Project Lead & AI Developer

NVIDIA/Viettel/Sovico

June 2, 2026 – June 12, 2026

- **Multi-Agent System Architecture:** Spearheaded the development of an enterprise-focused AI system utilizing specialized agents to analyze weather-related risks, delivering context-aware, actionable recommendations for sectors like tourism, construction, and agriculture.
- **Advanced LLMs & HPC Infrastructure:** Engineered a highly robust, containerized AI pipeline leveraging Qwen 3.5-72B (vLLM) and Nemotron 3 Super 120B (NVIDIA NIM), successfully deployed and powered by an NVIDIA 8x H200 GPU cluster.
- **Backend & Geospatial Intelligence:** Architected a high-performance data layer with FastAPI, PostgreSQL/PostGIS, Redis, and Qdrant (Vector DB), integrating complex spatial data (Overpass OSM, Apify) and real-time weather APIs (Earth2Studio, Open-Meteo).
- **Real-Time Streaming Interface:** Built a responsive Next.js Progressive Web App (PWA) incorporating WebSockets to ensure seamless, real-time delivery of weather intelligence and multi-agent communications to end-users.

Team Datdy - VinUni Datathon The GridBreakers 2026

Data Analyst

VinUni

Apr 18, 2026 – May 11, 2026

- **Complex Data Processing & Structuring:** Engineered robust data pipelines in Jupyter Notebooks utilizing Pandas and NumPy to clean and integrate a decade-long (2012–2022) e-commerce dataset, efficiently navigating 15 CSV files distributed across Master, Transaction, Analytical, and Operational layers.
- **Advanced EDA & Visualization:** Leveraged Matplotlib and Seaborn for comprehensive EDA, transforming multidimensional data into clear visualizations to reveal key trends in inventory, promotions, and web traffic.
- **Business Intelligence & Strategy:** Partnered with the HCMUT team to translate technical insights into actionable operational strategies, directly solving core business challenges for a simulated fashion retailer.

Morphysics - Bach Khoa Innovation 2026

Project Lead & Frontend Developer

HCMUT

Mar 2026 – Present

- **Interactive 2D Engine:** Architected a React 19/TypeScript virtual lab, integrating Matter.js to render real-time mechanics (collisions, gravity) with seamless drag-and-drop.
- **High-Performance UI:** Engineered a 60 FPS Glassmorphism Telemetry Dashboard via `requestAnimationFrame` to monitor live physical properties with minimal CPU/GPU overhead.
- **AI Chatbot & Architecture:** Implemented a multimodal AI UI executing API-driven simulations in <3s. Led the BKI pitch and built a modular 20-experiment library enforcing clean-code standards.

PROJECTS

BKAi — Agentic RAG Admissions System

Full-Stack AI Developer

Jan 2026 – Apr 2026

- **Autonomous Multi-Agent RAG:** Architected a 100% local workflow utilizing LangGraph and Ollama (Qwen2.5-7B). Engineered a self-reflection loop to cross-evaluate context, completely eliminating LLM hallucinations and achieving 100% grounded accuracy on admissions data.
- **Hybrid Search Pipeline:** Developed an advanced retrieval system fusing Vector (ChromaDB) and Lexical Search (BM25) via Reciprocal Rank Fusion (RRF). Integrated a Cross-Encoder reranker to maximize document ranking precision for highly specific, domain-centric queries.
- **Semantic Caching & Optimization:** Implemented a Redis-backed semantic cache using Sentence-Transformers, identifying query similarities to slash response latency by 99% (from ~90s to <0.1s). Built an automated data ingestion pipeline with smart chunking to prevent Out-of-Memory (OOM) errors.
- **Full-Stack Deployment & Voice UI:** Containerized a FastAPI/Nginx architecture via Docker Compose with WebSockets for real-time streaming. Integrated local *faster-whisper* and *edge-tts* for a natural Vietnamese voice interface, optimizing hardware usage to stably support 15 concurrent users.

YourAI — AI Assistant & Academic Management Platform

Full-Stack AI Developer

Feb 2026 – Present

- **Enterprise PWA Architecture:** Architected a Monorepo academic management platform using FastAPI and React (Vite). Optimized as a Progressive Web App (PWA), achieving a >90/100 Google Lighthouse score for instant load times and seamless cross-device installation.
- **AI Agent & NLP Processing:** Integrated Gemini 1.5 Flash via Function Calling to translate natural language into automated database commands with <1.2s latency. Engineered a Regex Fallback Parser to guarantee 99.9% system availability and 100% task success during API outages.
- **Dual-Scale Engine & Async Queues:** Built a real-time GPA engine converting Vietnamese (10-point) to Australian (7-point) scales. Implemented asynchronous queues via ARQ, Redis, and Resend SMTP for bulk email dispatching, boosting backend concurrency by 300% using `asyncpg`.
- **Zero-Trust Security & Infrastructure:** Enforced strict multi-tenant data isolation using Supabase PostgreSQL Row Level Security (RLS) and JWT. Developed a secure 60s OTP flow with bcrypt hashing and anti-brute force throttling on a highly optimized, zero-cost serverless stack.

CERTIFICATIONS

Certificates:

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| • IELTS: 7.0 | British Council |
| • Artificial Intelligence Fundamentals | IBM |
| • Generative AI with Large Language Models | DeepLearning.AI & AWS |
| • Intro to Machine Learning | Kaggle |
| • Intermediate Machine Learning | Kaggle |

TECHNICAL SKILLS

AI & Machine Learning: LangChain, LangGraph, MCP, Ollama (Qwen, Llama), Nemotron 3
Backend & Databases: Python, FastAPI, Node.js, REST APIs, Redis, ChromaDB, Qdrant
Frontend Development: React, Vite, HTML/CSS, Tailwind CSS, Chart.js
Data Analytics: Pandas, NumPy, Matplotlib, Seaborn
Tools & Infrastructure: Git, WebSockets, Nemo Agent Toolkit, VS Code, Antigravity